Practical # 2: Maven +Docker Integration

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8037]
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C:\Users\narut>mkdir docker-maven2

C:\Users\narut>cd docker-maven2
```

Step-01: Create a folder and get into that folder

```
C:\Users\narut\docker-maven2>mvn archetype:generate -DgroupId=com.example -DartifactId=docker-demo -DarchetypeArtifactId=maven-archetype-quickstart -Dintera
ctiveMode=false
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/apache-maven-3.9.11/lib/guice-5.1.0
-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> archetype:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< archetype:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- archetype:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: C:\Users\narut\docker-maven2
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: docker-demo
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: C:\Users\narut\docker-maven2\docker-demo
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.404 s
[INFO] Finished at: 2026-03-26T12:37:16+05:30
[INFO] ------------------------------------------------------------------------

C:\Users\narut\docker-maven2>cd docker-demo

C:\Users\narut\docker-maven2\docker-demo>dir
```

STEP-02: Create maven project

mvn archetype:generate -DgroupId=com.example -DartifactId=docker-demo -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

```
C:\Users\narut\docker-maven2>cd docker-demo

C:\Users\narut\docker-maven2\docker-demo>dir
 Volume in drive C is Windows
 Volume Serial Number is 2461-2C7D

 Directory of C:\Users\narut\docker-maven2\docker-demo

26-03-2026  12:37    <DIR>          .
26-03-2026  12:37    <DIR>          ..
26-03-2026  12:37               664 pom.xml
26-03-2026  12:37    <DIR>          src
               1 File(s)            664 bytes
               3 Dir(s)  14,787,354,624 bytes free

C:\Users\narut\docker-maven2\docker-demo>notepad pom.xml

C:\Users\narut\docker-maven2\docker-demo>
```

STEP-03: Get inside the project and check the directory

```
C:\Users\narut\docker-maven2\docker-demo>notepad pom.xml

C:\Users\narut\docker-maven2\docker-demo>notepad dockerfile

C:\Users\narut\docker-maven2\docker-demo>ren dockerfile.txt dockerfile
```

STEP-04: Make changes in the pom.xml, create a dockerfile and then rename it.

<project xmlns="http://maven.apache.org/POM/4.0.0"

xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

xsi:schemaLocation="http://maven.apache.org/POM/4.0.0

< http://maven.apache.org/xsd/maven-4.0.0.xsd">

```
<modelVersion>4.0.0</modelVersion>
  <!-- Must match your archetype command -->
  <groupId>com.example</groupId>
  <artifactId>docker-demo</artifactId>
  <version>1.0-SNAPSHOT</version>
  <packaging>jar</packaging>
  <name>docker-demo</name>
  <!-- Java Version -->
  <properties>
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
    <maven.compiler.source>21</maven.compiler.source>
    <maven.compiler.target>21</maven.compiler.target>
  </properties>
  <!-- Dependencies -->
  <dependencies>
    <!-- JUnit -->
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>4.13.2</version>
      <scope>test</scope>
    </dependency>
  </dependencies>
  <!-- Build Plugins -->
  <build>
    <plugins>
      <!-- Executable JAR -->
      <plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-jar-plugin</artifactId>
        <version>3.3.0</version>
        <configuration>
          <archive>
            <manifest>
              <mainClass>com.example.App</mainClass>
            </manifest>
          </archive>
```

```
            </configuration>
        </plugin>

        <!-- Docker Plugin -->
        <plugin>
            <groupId>com.spotify</groupId>
            <artifactId>dockerfile-maven-plugin</artifactId>
            <version>1.4.13</version>

            <configuration>
                <!-- Must match project name -->
                <repository>docker-demo</repository>
                <tag>${project.version}</tag>
                <!-- Important -->
                <dockerfile>${project.basedir}/Dockerfile</dockerfile>
            </configuration>
            <executions>
                <execution>
                    <id>default</id>
                    <goals>
                        <goal>build</goal>
                    </goals>
                </execution>
            </executions>
        </plugin>
    </plugins>
  </build>
</project>
```

Dockerfile

```
FROM eclipse-temurin:21-jdk
WORKDIR /app
COPY target/docker-demo-1.0-SNAPSHOT.jar app.jar
ENTRYPOINT [&quot;java&quot;,&quot;-jar&quot;,&quot;app.jar&quot;]
```

```
C:\Users\narut\docker-maven2\docker-demo>cd src

C:\Users\narut\docker-maven2\docker-demo\src>cd main

C:\Users\narut\docker-maven2\docker-demo\src\main>cd java

C:\Users\narut\docker-maven2\docker-demo\src\main\java>cd com

C:\Users\narut\docker-maven2\docker-demo\src\main\java\com>cd example

C:\Users\narut\docker-maven2\docker-demo\src\main\java\com\example>cd App.java
The directory name is invalid.

C:\Users\narut\docker-maven2\docker-demo\src\main\java\com\example>notepad App.java

C:\Users\narut\docker-maven2\docker-demo\src\main\java\com\example>cd ..

C:\Users\narut\docker-maven2\docker-demo\src\main\java\com>cd..

C:\Users\narut\docker-maven2\docker-demo\src\main\java>cd ..

C:\Users\narut\docker-maven2\docker-demo\src\main>cd ..

C:\Users\narut\docker-maven2\docker-demo\src>cd ..

C:\Users\narut\docker-maven2\docker-demo>mvn clean package
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/apache-maven-3.9.11/lib/guice-5.1.0
-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< com.example:docker-demo >-----------------------
[INFO] Building docker-demo 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/com/spotify/dockerfile-maven-plugin/1.4.13/dockerfile-maven-plugin-1.4.13.pom
Downloaded from central: https://repo.maven.apache.org/maven2/com/spotify/dockerfile-maven-plugin/1.4.13/dockerfile-maven-plugin-1.4.13.pom (5.6 kB at 15 kB
/s)
```

STEP-05: Modify Java Program with path : src/main/java/com/example/App.java

STEP-06: Build the package

```
Command Prompt

[INFO]  ---> d552edd869aa
[INFO] Successfully built d552edd869aa
[INFO] Successfully tagged docker-demo:1.0-SNAPSHOT
[INFO]
[INFO] Detected build of image with id d552edd869aa
[INFO] Building jar: C:\Users\narut\docker-maven2\docker-demo\target\docker-demo-1.0-SNAPSHOT-docker-info.jar
[INFO] Successfully built docker-demo:1.0-SNAPSHOT
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  31.998 s
[INFO] Finished at: 2026-03-27T01:04:10+05:30
[INFO] ------------------------------------------------------------------------

C:\Users\narut\docker-maven2\docker-demo>docker run docker-demo:1.0-SNAPSHOT
Hello from Docker Maven Demo!

C:\Users\narut\docker-maven2\docker-demo>docker images
                                                                                                                                  Info →
  U  In Use
IMAGE                                    ID             DISK USAGE   CONTENT SIZE   EXTRA
alpine:latest                            25109184c71b       13.1MB         3.95MB    U
ashika0804/ab2:v1.0                      28e750845649        476MB          148MB
ashika0804/ap1:v1                        5c1c79cf906a       33.6MB         10.1MB
ashika0804/c1:v1.1                       cd1dba651b30        119MB         31.7MB    U
docker-demo:1.0-SNAPSHOT                 d552edd869aa        675MB          211MB    U
docker/welcome-to-docker:latest          c4d56c24da4f       22.2MB         6.03MB    U
eclipse-temurin:21-jdk                   3b0a98dfbdf1        679MB          214MB
ghcr.io/ashika07734/ghcr-demo:latest     25d6eacf3081       92.5MB           26MB
ghcr.io/ashika07734/sql_7739:v1          45abbc49f01a        784MB          166MB    U
httpd:latest                             dd178595edd6        177MB         47.6MB    U
mongo:6                                  03cda579c8ca       1.06GB          273MB    U
mysql:5.7                                4bc6bc963e6d        700MB          149MB
mysql:8.0                                aa956cd6c5fa        784MB          166MB    U
mysql:8.1                                a1361596e618        784MB          166MB    U
mysql:latest                             6b18d01fb632       1.27GB          283MB    U
nginx:latest                             c881927c4077        240MB         65.7MB    U
node-mongo-compose-app:latest            a6292355b97b       1.63GB          406MB    U
```

STEP-07: Run the container

STEP-08: Verify the image